Informational Actualization Model

Results Inventory & Paper Reference Map

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https://github.com/hmahaffeyges/IAM-Validation
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Core premise: IAM adds a single Hubble friction term $\beta_m E(a)$ to the matter perturbation equations only, where $\beta_m = \Omega_m/2$ is derived from the virial theorem and $E(a) = \exp(1-1/a)$ from horizon thermodynamics. Background ΛCDM is unchanged. Zero free parameters beyond the standard model.

Two root discoveries: (1) The thermodynamic identity $K = Q = T\Delta S = E_{\mathrm{Landauer}}$ governing the virial theorem. (2) The informational addition to the Jacobson entropy functional. Everything in this document is a consequence applied consistently.

Contents

Category I — The Cosmological Model

Perturbation-level modification to ΛCDM, tested against real data. One mechanism, zero new free parameters, addresses multiple open tensions simultaneously.

The σ_8 Tension

Structure growth surveys have consistently found less clustering than ΛCDM predicts for over a decade. No standard extension resolves this without adding free parameters or sacrificing fits elsewhere.

- IAM predicts $\sigma_8 = 0.800$ from first principles via $\mu(a) < 1$, suppressing matter growth without touching photon propagation.
- KiDS-Legacy + DES Y3 + DESI joint analysis (2025) finds $\sigma_8 = 0.802 \pm 0.020$. Consistent at 0.1σ.
- No adjustment was made to chase this number. The prediction precedes the result and is timestamped in the repository.

[20] Theory Paper · [22] Constraints on Late-Time $f\sigma_8$ Suppression · [4] Dual-Sector Perturbation Cosmology: IAM–CAMB · [24] The Redshift-Dependent S_8 Trend in the Context of IAM

The Hubble Tension

Local distance ladder finds $H_0 \approx 73$ km/s/Mpc; Planck CMB finds $H_0 \approx 67$ km/s/Mpc. Eighteen years of effort have not resolved it. Every proposed solution addresses one side, not both simultaneously.

- IAM derives a physical reason for the split: photons travel null geodesics and do not decohere; matter travels timelike worldlines and does. They experience different effective gravitational coupling.
- $H_0(\text{photon}) = 67.16 \pm 0.47$ km/s/Mpc (0.37σ from Planck).
- $H_0(\text{matter}) = 72.26$ km/s/Mpc (0.75σ from SH0ES). Both tensions addressed simultaneously. One mechanism. Zero new parameters.

[20] Theory Paper · [4] Dual-Sector Perturbation Cosmology: IAM–CAMB · [5] Dual-Sector Validation Paper · [11] IAM Dual-Sector Note

The $f\sigma_8$ Tension

Galaxy growth rate measurements consistently fall below ΛCDM predictions across multiple independent surveys spanning a decade.

- IAM predicts a specific suppression curve $f\sigma_8(z)$ governed by $E(a) = \exp(1 - 1/a)$. Suppression peaks at 7–8% near $z \approx 0.3$ and decays to zero at high z, where IAM recovers ΛCDM exactly. This shape is not fitted — it is what $E(a)$ produces.
- Confronted against DESI DR1 full-shape measurements at six redshift bins ($0.295 \leq z_{\text{eff}} \leq 1.491$): consistent throughout.
- Recovery to ΛCDM at high z is a prediction, not a vulnerability. IAM is not suppressed where it should not be.

[22] Constraints on Late-Time $f\sigma_8$ Suppression · [3] Dark Energy or Sector Tension?

The Phantom Crossing

DESI reported a preference for dynamical dark energy with an apparent phantom crossing at $z \approx 0.3$–0.4, interpreted as possible new physics beyond ΛCDM.

- IAM explains this without requiring dynamical dark energy. A single-fluid w_0w_aCDM fitter ingesting ΛCDM-consistent BAO distances alongside suppressed matter growth is forced to infer $w_{\text{eff}}(z) \neq -1$ to reconcile them.
- The phantom crossing is a sector-assignment artifact, not a signal of new dark energy physics.
- The DESI DR2 phantom crossing redshifts ($z_{\text{cross}} = 0.33$–0.43) coincide exactly with the redshift range where IAM growth suppression is most prominent.

[3] Dark Energy or Sector Tension? IAM Confrontation with DESI Full-Shape Growth Rates and Joint Weak Lensing

Core MCMC Results and Decisive Test

- $\beta_m = \Omega_m/2 = 0.1575$ derived from the virial theorem. Planck 2018 MCMC posterior: 0.1583 ± 0.0033. Consistent at 0.2σ. Not fitted.
- 17 independent MCMC chains against the full Planck 2018 likelihood. All 17 converged. None discarded. $R-1 < 0.01$.
- Growth-geometry split: DESI DR1 FS+BAO finds $\Omega_m^{\text{growth}} = 0.2962 \pm 0.0095$; IAM predicts 0.2952 at $z \sim 0.5$. Consistent at 0.02σ.
- Transition redshift: ΛCDM predicts $z_t = 0.632$; IAM predicts $z_t = 0.718$. Three independent published compilations find 0.698–0.740. IAM inside all three; ΛCDM below all three.
- $\Delta\chi^2(\text{IAM} - \Lambda\text{CDM}) = +0.54$. IAM fits the data as well as ΛCDM while addressing all tensions with zero new parameters.
- **Decisive test:** $\mu_0 = -0.136$ predicted. Euclid DR1 October 2026: projected 3.4σ sensitivity. On record.

[20] Theory Paper · [4] Dual-Sector Perturbation Cosmology: IAM–CAMB · [10] IAM–CAMB Technical Note · [23] Supplementary Methods & Reproducibility Guide · [19] Complete Test Validation Compendium · [18] Falsifiable Predictions of IAM · [16] IAM Official Score Card · [17] IAM Overview Companion · [14] IAM Master Consolidated Preprint

Category II — The Thermodynamic Identity

Stands completely alone. Requires no cosmology, no IAM, no MCMC. Verifiable by any physicist in any field using only classical mechanics and thermodynamics.

The virial theorem has been verified across 37 orders of magnitude since Clausius in 1870. In 154 years one question was never answered: what physically *is* K?

- **The identity:** $K = Q = T\Delta S = E_{\text{Landauer}}$. The time-averaged kinetic energy in the virial theorem is identically equal to the Landauer cost of virialization.

- Derived from four things only: the first law, the second law, Landauer's principle, and the equilibrium condition of the $1/r$ potential. No additional assumptions.
- The virial temperature cancels exactly, making the identity scale-invariant. Holds at the scale of a hydrogen atom and a galaxy cluster by the same mathematics.
- Verified from hydrogen atoms (10^{-10} m) to the cosmological coupling constant (10^{26} m): 37 orders of magnitude.
- The virial theorem is the *mechanical expression* of this thermodynamic identity. Clausius derived the mechanical statement. The thermodynamic content was always there, unrecognized.

The $n = 5/2$ Bridge: Quantum to Cosmological

This is the precise junction where quantum decoherence and cosmological structure formation meet in a single derived number, arrived at from two completely independent directions.

Top-down (from the cosmological activation function): $E(a) = \exp(1 - 1/a)$ is derived from horizon thermodynamics. For its cumulative integral to yield the required $1/a$ dependence in the exponent, the instantaneous decoherence rate must scale as $D(a)^n$ where the matter-domination algebra requires:

$$n - \tfrac{9}{2} = -1 \quad \Longrightarrow \quad \boxed{n = \tfrac{5}{2}} \tag{1}$$

The functional form of $E(a)$ demands this exponent from the top down.

Bottom-up (from the physics of decoherence at structure formation scales): linear perturbations give $n = 1$; fully nonlinear collapse gives $n \gtrsim 4$. The Press-Schechter threshold collapse regime — where most information is produced — sits between them. Six independent N-body simulation groups measure $n_{\text{eff}} = 3.22 \pm 0.44$, with the analytical matter-domination value $n = 5/2$ at the lower bound and the full ΛCDM history shifting it to 3.0–4.0 as expected.

Two independent routes — one asking what the cosmological activation function requires from the top, the other asking what quantum decoherence at structure formation scales produces from the bottom — return the same number. This is the framework closing on itself across 26 orders of magnitude. It is the precise, quantitative connection between quantum decoherence events and cosmological structure.

[36] The Thermodynamic Identity Governing the Virial Theorem: Derivation from First Principles and Evidence Across 37 Orders of Magnitude · [26] The Virial Partition Across Wide Range of Physical Scales · [25] Virial Efficiency and Effective Nonlinear Exponent in IAM · [20] Theory Paper · [2] A Note on Entropic Gravity

Category III — The Cosmological Constant

The worst prediction in the history of physics — addressed from first principles with zero free parameters.

Quantum field theory predicts a vacuum energy 10^{122} times larger than the observed cosmological constant. This is the largest discrepancy between theory and observation in all of science. Every prior serious attempt either introduces free parameters, invokes fine-tuning, or resorts to anthropic reasoning.

- **Result:** $\Lambda/\rho_{\text{vac}} = (2/\pi)(l_P/l_H)^2 \times (\Omega_b/\Omega_m) = 1.38 \times 10^{-123}$ vs. observed 1.14×10^{-123}. Factor of 1.2. Zero free parameters.

- **Closes 121.97 of 122 orders of magnitude** — not by cancellation or fine-tuning, but by physical identification of what vacuum energy and the cosmological constant actually are.
- **Physical identification:** vacuum energy = total Planck-scale potential (pure potency, not yet actualized). Cosmological constant = accumulated Landauer cost of baryonic decoherence (accumulated actual).
- **The Ω_b/Ω_m factor is physically necessary, not fitted.** Dark matter is already classical geometry and does not independently decohere. Only baryons contribute. Remove this factor and the derivation fails on physical grounds.
- **The $2/\pi$ factor is exact in the de Sitter limit.** Each observer's causal patch subtends exactly half the full boundary. Calculable, not assumed.
- **Remaining factor of** 1.2 has two identified calculable sources: departure from exact de Sitter today and the full cosmic decoherence history integral. No free parameters introduced.
- **Secondary prediction:** $w > -1$. Consistent with DESI DR2.

[30] The Cosmological Constant as Actualized Vacuum Energy

Category IV — Time and the Arrow of Time

Resolves one of the deepest foundational problems in physics from first principles. The arrow of time is not a statistical artifact — it is a thermodynamic necessity with a specific physical origin.

- **Three distinct quantities identified** that are routinely conflated: (i) coordinate time — a redundant label, no preferred direction; (ii) proper time τ — geometric path length, reversible in principle; (iii) duration — measured by $E(a)$, thermodynamic, irreversible, with a specific physical beginning.
- **Duration has a beginning:** the electroweak transition at $t = 10^{-12}$ s. Before electroweak symmetry breaking, no massive particles exist and $E(a)$ cannot be nonzero. Duration begins when mass begins. The Higgs boson is the origin of duration, the first irreversible act.
- **Duration has a direction:** each decoherence event writes information permanently on the cosmic horizon. $E(a)$ is monotonically increasing by construction. Potential moves toward act, never the reverse. The arrow of time is a thermodynamic necessity, not a statistical artifact of initial conditions.
- **Duration accumulates on massive particles only.** Photons travel null geodesics and accumulate neither proper time nor duration in the thermodynamic sense. This is the physical origin of the sector split that drives every IAM cosmological prediction.
- **Duration has a limit:** $E(\infty) = e \approx 2.718$. The universe approaches but never reaches full actualization. Asymptotically complete, never finished.

[32] The Two Faces of Time: Coordinate Time, Proper Time, and Duration · [29] The Higgs Boson and the Origin of Duration

Category V — The Dark Sector Identification

Dark matter and dark energy are not two mysteries. They are two sides of one equation operating at cosmological scales.

- **Dark matter = geometric potential half of the virial partition.** At each gravitational decoherence event, half the energy deposits into spacetime curvature. Stored geometrically. Does not dilute. This is what observational cosmology calls dark matter.
- **Dark energy = kinetic informational half of the virial partition.** The other half crosses into classical actuality as Landauer pressure on the cosmological horizon.
- These are two expressions of $2K + V = 0$ at cosmological scales, not two independent phenomena.
- **The coincidence problem is dissolved.** The virial ratio $\Omega_m a^{-3}/[\beta_m E(a)]$ descends from ~ 16 million at $z = 9$ to exactly 2.0 at $a = 1$. Not fine-tuning. A derivation. The universe reaches virial equilibrium now because $\beta_m = \Omega_m/2$ by derivation and $E(1) = 1$ by thermodynamic construction.
- **The anthropic selection and the physics point to the same moment.** The physics explains the timing without invoking observers to explain the physics.

> *[28] Dark Matter and Dark Energy as Virial Partners · [34] The Boundary Between Potential and Actual: Gravitational Decoherence, the Virial Partition, and the Emergence of Classical Structure · [27] Dark Energy Evolution and the Far Future of an IAM Universe*

Category VI — Black Holes and the M–σ Relation

A 25-year-old empirical scaling law derived from first principles, with the same coupling constant that governs cosmological structure.

- **Result:** $M_{\rm BH} = 2.00 \times 10^8 \times (\sigma/200\,{\rm km/s})^4\, M_\odot$. Derived, not fitted. Matches observations to 2%.
- **Normalization** $f_{\rm geom} = \Omega_m/(2\pi)$ derived from IAM coupling structure. Same coupling constant governing cosmological structure determines black hole masses.
- **Black holes in IAM are mandatory local encoding surfaces.** They form when the local decoherence rate saturates the Bekenstein–Hawking bound. The M–σ relation is a consequence of that saturation condition.
- **Universal Landauer identity:** $E_{\rm Landauer} = (\ln 2/2) \times M_{\rm BH}c^2$. Mass-independent. Same encoding cost per bit for stellar-mass and supermassive black holes.
- **Two-horizon framework:** black hole horizon (hot, local encoding) feeds to cosmic horizon (cold, global encoding).
- **Information paradox reframed:** matter does not enter a black hole in IAM — the projection ceases. The particle exists as a stable decoherence loop on the cosmic holographic boundary.

[13] The M–σ Relation from Gravitational Decoherence Thermodynamics · [8] Black Hole Horizons as Thermodynamic Encoding Surfaces · [9] The Cessation of Projection: Why Matter Does Not Enter a Black Hole and What This Means for the Information Paradox · [1] Three-Way Mass Discrepancy in Galaxy Clusters · [12] Lensing-Dynamics Mass Discrepancy as a Redshift-Dependent Test

Category VII — Quantum Foundations

IAM's thermodynamic chain connects to foundational questions in quantum mechanics, measurement theory, and quantum Darwinism.

- **Measurement problem dissolved:** the quantum-to-classical transition is not an interpretive puzzle in IAM — it is a thermodynamic event. Each decoherence event has a Landauer cost and produces a permanent record on the horizon. The same mechanism drives cosmological structure formation.
- **Quantum Darwinism at cosmological scales:** the cosmic horizon functions as the environment that selects pointer states. Classical structure emerges because only the decoherent branch survives encoding. The cosmological horizon is the ultimate redundant record.
- **Gravitational decoherence at the quantum level:** the decoherence rate from gravitational structure formation is derived from first principles and connected to the Lindblad master equation framework.

[15] The Measurement Problem Dissolved · [33] Quantum Darwinism at Cosmological Scales: Gravitational Decoherence, the Cosmic Horizon, and the Emergence of Classical Structure · [7] Gravitational Decoherence from Dual-Sector Thermodynamics

Category VIII — Particle Physics

Conditional results — novel physical motivation, derivations not yet complete. Stated as theorems and predictions, not established results.

- **Electron rest mass as a fixed point of gravitational decoherence.** The electron mass is the scale at which the decoherence rate equals the Compton frequency. Not fitted.
- **Koide relation.** The mysterious numerical relation between electron, muon, and tau masses (accurate to 9 significant figures) approached via the virial theorem route — physical motivation not previously taken. Stated as a conditional theorem. Known weaknesses identified and documented.
- **Baryon asymmetry.** $\eta \sim 6 \times 10^{-10}$ predicted from the information writing constraint. MCMC float test pending.

[6] Electron Rest Mass from Holographic Horizon Thermodynamics · [21] Three Charged Lepton Generations and the Koide Ratio from Holographic Boundary Encoding · [31] Matter-Antimatter Asymmetry and the Information Writing Constraint

Category IX — Structure Formation Predictions

Predictions made and timestamped in the GitHub repository. Not fits to data.

- **Missing satellites problem.** ΛCDM predicts far more small satellite galaxies than observed. IAM predicts a minimum virialization scale from the Bekenstein–Hawking saturation condition. Minimum halo mass calculable from the saturation condition.
- **Cusp-core problem.** Simulations predict dense cusps; observations find flat cores. IAM derives $r_{\text{core}} \sim \sigma^2$ exactly from the $E(a)$ actualization clock. Scaling exact. The $\sim 130\times$ normalization offset is identified as the ratio of local black hole encoding rate to cosmic rate. Open problem, not a failure.
- **Globular cluster age tension.** IAM predicts a 1.378 Gyr excess from the actualization clock. Valcin+2025 finds 1.397 Gyr. Consistent at 0.10σ. Timestamped March 2026.
- **Axis of evil.** IAM predicts $A_{\text{IAM}} = 0.0628$ from the de Sitter causal boundary. Observed: 0.066 ± 0.021. Consistent at 0.15σ. Timestamped March 2026.

[35] Missing Satellites: The Virial Partition Closure Condition and the Two Mechanisms · [18] Falsifiable Predictions of IAM · [19] Complete Test Validation Compendium

Summary Table

Category	**Key Result**	**Status**
I. Cosmological	$\sigma_8 = 0.800$, H_0 sector split	17 chains, all converged
I. Cosmological	$f\sigma_8$ suppression, phantom crossing	Consistent DESI DR1/DR2
I. Cosmological	$\mu_0 = -0.136$ predicted	Euclid DR1 Oct 2026
II. Thermo Identity	$K = Q = T\Delta S = E_{\text{Landauer}}$	Verified 37 orders of magnitude
II. Thermo Identity	$n = 5/2$ from two directions	Quantum-to-cosmological bridge
III. Cosm. Const.	Closes 121.97/122 orders	Zero free parameters
IV. Arrow of Time	Duration derived, origin identified	Sector split explained
V. Dark Sector	DM and DE as virial partners	Coincidence problem dissolved
VI. Black Holes	M–σ derived, 2% match	Zero free parameters
VII. Quantum Found.	Measurement problem, Darwinism	Thermodynamic derivation
VIII. Particle Phys.	Electron mass, Koide, baryon η	Conditional / pending
IX. Structure Form.	Missing sats, cusp-core, age, axis	Timestamped predictions

Complete Paper Reference List

All 36 active papers as of March 2026. Numbers in brackets throughout this document refer to this list.

#	Title	Category
1	Three-Way Mass Discrepancy in Galaxy Clusters as a Test of $\mu < 1$, $\Sigma = 1$	Cat. VI
2	A Note on Entropic Gravity and the Thermodynamic-Gravity Conjecture	Cat. II
3	Dark Energy or Sector Tension? IAM Confrontation with DESI Full-Shape Growth Rates and Joint Weak Lensing	Cat. I
4	Dual-Sector Perturbation Cosmology: IAM–CAMB Implementation *(Level 2)*	Cat. I
5	Dual-Sector Expansion: Type Ia Supernovae Validate Matter-Sector H_0 Normalization with ΛCDM Geometric Consistency	Cat. I
6	Electron Rest Mass from Holographic Horizon Thermodynamics	Cat. VIII
7	Gravitational Decoherence from Dual-Sector Thermodynamics	Cat. VII
8	Black Hole Horizons as Thermodynamic Encoding Surfaces	Cat. VI
9	The Cessation of Projection: Why Matter Does Not Enter a Black Hole, and What This Means for the Information Paradox	Cat. VI
10	IAM–CAMB Technical Note	Cat. I
11	On the Dual-Sector Structure of IAM	Cat. I
12	Lensing-Dynamics Mass Discrepancy as a Redshift-Dependent Test of IAM	Cat. VI
13	The M–σ Relation from Gravitational Decoherence Thermodynamics	Cat. VI
14	IAM Master Consolidated Preprint	Overview
15	The Measurement Problem Dissolved	Cat. VII
16	IAM Official Score Card	Cat. I
17	IAM Overview Companion	Overview
18	Falsifiable Predictions of the Informational Actualization Model	Cat. I, IX
19	Informational Actualization Model: Complete Test Validation Compendium	Cat. I, IX
20	Horizon Thermodynamics and Gravitational Decoherence: Theoretical Foundations *(Core theory paper)*	All categories
21	Three Charged Lepton Generations and the Koide Ratio from Holographic Boundary Encoding	Cat. VIII
22	Constraints on Late-Time $f\sigma_8$ Suppression from $\mu < 1$, $\Sigma = 1$: Confrontation with Planck and Large-Scale Structure *(Level 1)*	Cat. I
23	Supplementary Methods & Reproducibility Guide	Methods
24	The Redshift-Dependent S_8 Trend in the Context of the Informational Actualization Model	Cat. I

#	Title	Category
25	Virial Efficiency and Effective Nonlinear Exponent in IAM: Published N-body Confirmation of $\beta_m = \Omega_m/2$	Cat. II
26	The Virial Partition Across Wide Range of Physical Scales: Cross-Domain Validation of IAM	Cat. II
27	Dark Energy Evolution and the Far Future of an IAM Universe	Cat. V
28	Dark Matter and Dark Energy as Virial Partners	Cat. V
29	The Higgs Boson and the Origin of Duration	Cat. IV
30	The Cosmological Constant as Actualized Vacuum Energy	Cat. III
31	Matter-Antimatter Asymmetry and the Information Writing Constraint	Cat. VIII
32	The Two Faces of Time: The Distinction Between Coordinate Time, Proper Time, and Duration	Cat. IV
33	Quantum Darwinism at Cosmological Scales: Gravitational Decoherence, the Cosmic Horizon, and the Emergence of Classical Structure	Cat. VII
34	The Boundary Between Potential and Actual: Gravitational Decoherence, the Virial Partition, and the Emergence of Classical Structure	Cat. V
35	Missing Satellites: The Virial Partition Closure Condition and the Two Mechanisms	Cat. IX
36	The Thermodynamic Identity Governing the Virial Theorem: Derivation from First Principles and Evidence Across 37 Orders of Magnitude	Cat. II

All results trace to two core discoveries: the thermodynamic identity $K = Q = T\Delta S = E_{\text{Landauer}}$, and the informational addition to the Jacobson entropy functional. Everything else is a consequence applied consistently. Repository timestamps establish priority for all predictions listed above.

GRF Essay (submitted February 2026): *Gravitational Decoherence on Timelike Worldlines: The Missing Information Was Always Information.* Results announced May 15, 2026.

Decisive observational test: Euclid DR1, October 2026. Predicted $\mu_0 = -0.136$ at projected 3.4σ sensitivity.